

Proposal for SiEPIC Design

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Abstract—In this report, I report the design details for the course of Silicon Photonic Integrated Circuits Design to provide a layout of Mach-Zehnder Interferometers (MZIs) to be fabricated in the next steps. The design process includes the mode analysis of the waveguides (for 450 nm waveguides) and layout designs based on the process design kit (PDK) provided for the SiEPIC process, based on the electron beam lithography. To this end, four MZI devices are designed based on y-branches with diverse waveguide length differences, for both the transverse electric and magnetic (TE and TM) modes. Moreover, two identical TE devices comprise broadband couplers at one end to obtain insights into the effects of fabrication variation. Furthermore, two TM devices include adiabatic couplers with different lengths. The results of the simulations performed by the compact model library provided for the SiEPIC PDK are reported for next-step analysis in the design-fabrication-

Index Terms— Waveguide Modes, Layout Design, Mach-Zehnder Interferometer, Compact Model, Process Design Kit.

I. INTRODUCTION

AS one of the fundamental building blocks applicable in various kinds of photonic integrated circuits (PICs), it seems a good start to get hands on experience, starting with a Mach-Zehnder Interferometer (MZI). Therefore, in this proposal, as suggested by the instructors, I include MZIs.

II. DEVICE DESIGN

A. Waveguide Modes

For this design, a waveguide with 450 nm width on a standard silicon on insulator (SOI) wafer has been chosen (with a silicon thickness of 220 nm). Fig. 1(a) shows the electric field norm on a logarithmic scale, ensuring that the simulation area is large enough for a reliable simulation, whereas the linear electric field magnitude is shown in Fig. 2(b) for a transverse electric (TE) waveguide mode. Fig. 3(c) and (d) also show the $\log|E|$ and $|E|$ for the TM modes.

The effective refractive index at the 1.55 μm for this waveguide's TE and TM modes is $n_{\text{eff,TE}} = 2.357$ and $n_{\text{eff,TM}} = 1.732$, respectively. However, the investigation of the simulation environment showed that a distance of $\lambda_0 / 2 n_{\text{SiO}_2}$ from the waveguide's walls is enough to have a field damping of 10^{-4} with respect to the $|E|$ maximum value for TE mode. For the TM mode, on the other hand, a $\lambda_0 / n_{\text{SiO}_2}$ is needed (which is twice the one needed for the TE mode). This arises from both the mode shape and the less mode's confinement. Therefore, when working with the TM mode, one should carefully consider a sufficiently large simulation area. Furthermore, this

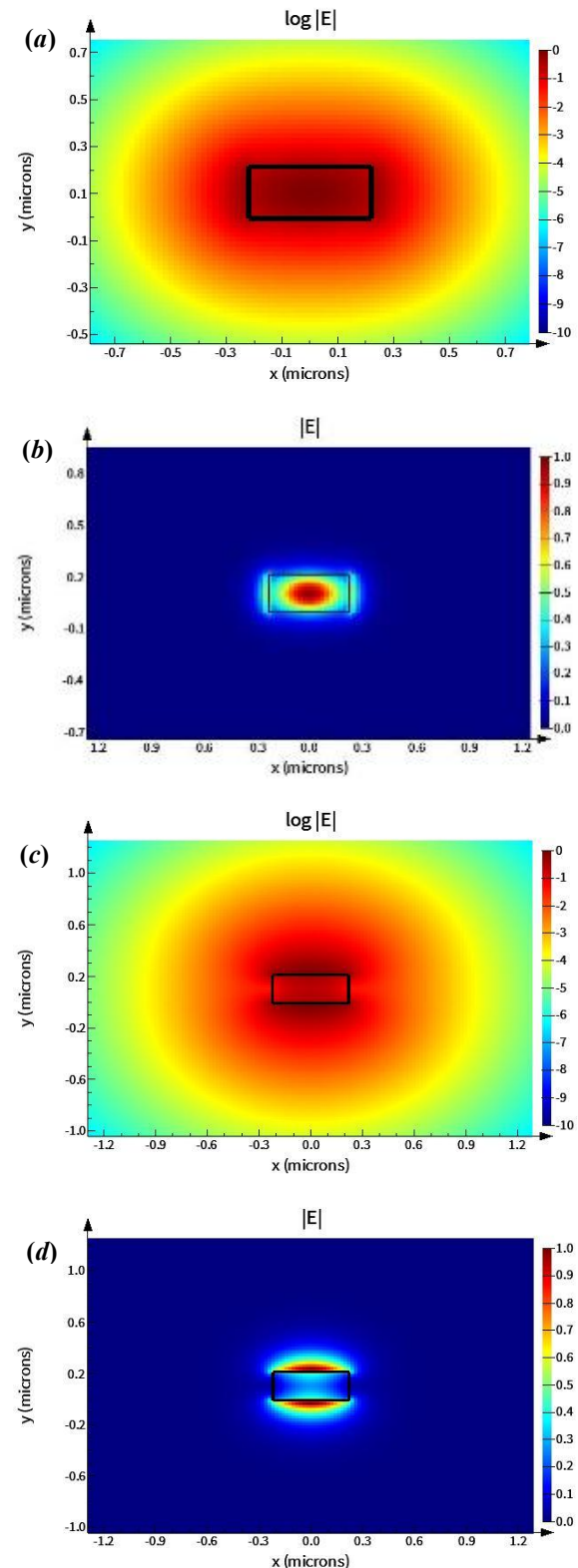


Fig. 1. a) The logarithmic and b) the linear magnitude of the electrical field.

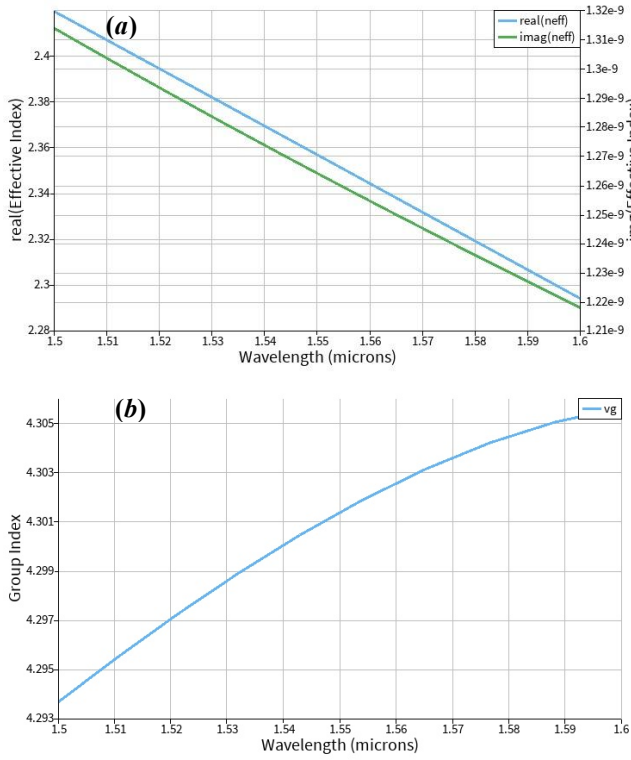


Fig. 2. a) Real and imaginary parts of the effective index and b) group index of the 450 nm width strip waveguide.

demonstrates that, for sensing applications, the TM mode could be more sensitive to the medium under recognition, especially when the analyte interacts with the top surface of the waveguides. However, the effective refractive indices reported here are for a big enough simulation region ($\sim 5 \times 5 \mu\text{m}^2$) where a fine mesh ($20 \times 20 \text{ nm}^2$ mesh elements) was applied. At $\lambda_0 = 1.55 \mu\text{m}$, the group indices of the TE and TM modes are 4.301 and 3.619, respectively.

The following equation represents a compact model based on Taylor's expansion around the central wavelength of $\lambda_0 = 1.55 \mu\text{m}$, for the simulated strip waveguide with 450 nm width for the TE₀ Mode, and its refractive and group indices are sketched in Fig. 2.

$$n_1 + n_2 (\lambda - \lambda_0) + n_3 (\lambda - \lambda_0)^2 = 2.35698 + 1.25417 (\lambda - \lambda_0) - 0.0396554 (\lambda - \lambda_0)^2 \quad (1)$$

The initial purpose of this design was to design a structure based on 450 nm waveguides; however, since there is some uncertainty about coupling between the grating, broadband, and adiabatic couplers, as well as other components standardized for 500 nm waveguides, and there are no adiabatic taper couplers between 450 and 500 nm waveguides, we proceeded using standard 0.5 μm waveguides and structures.

B. Devices Layout Design

In the next step, the design focused on both TM and TE mode MZIs. To this end, according to the available floor plan ($605 \times 410 \mu\text{m}^2$), for a space-efficient design, two different scenarios were considered for each mode:

In TE mode, we considered four MZI using y-branches, where the waveguide differences ΔL are 22.5, 45, 67.5, and 225 μm . Additionally, two identical MZI devices were designed

using broadband couplers with $\Delta L = 208.5 \mu\text{m}$ to evaluate the fabrication variation to some extent, providing intuition about practical imperfections. It should be noted that the group index is calculated ~ 4.1961 around 1550 nm for this mode, and we predict to have at least 2 to a maximum of ~ 20 oscillations in 50 nm wavelength span.

In the TM mode, four MZI devices were designed, again based on y-branches with $\Delta L = 15, 30, 45$, and 130 μm . These length differences were chosen since $n_g = 3.72$ in TM mode, therefore, we can have one to 10 oscillations for the designed devices in the 50 nm wavelength bandwidth. Moreover, two devices were designed with $\Delta L = 60$ and 130 μm using adiabatic TM mode couplers.

For both the TE and TM modes, 150 μm waveguides were coupled to TE and TM grating couplers for de-embedding purposes, which could be helpful to achieve the insertion losses after the wafer is fabricated.

II. RESULTS

A. Simulation Results

The simulation results using the Lumerical Interconnect are reported in this section. The results for the four MZIs with y-branches and broadband couplers in TE modes are shown in Fig. 3, whereas the results for the TM mode are exhibited in Fig. 4. As can be seen from the results, the waveguides' length differences demonstrate the FSR. It should be noted that wherever couplers were used, a taper terminator is attached to the unused branch of the coupler for the designs presented here.

B. Fabrication Results

The fabrication results of the designs for both TE and TM mode MZIs were satisfactory. Here, we start with the TE mode results for $\Delta L = 130 \text{ nm}$, and then, tabularize the results to conclude the fabrication process variations on the waveguides n_g and the MZIs' FSR.

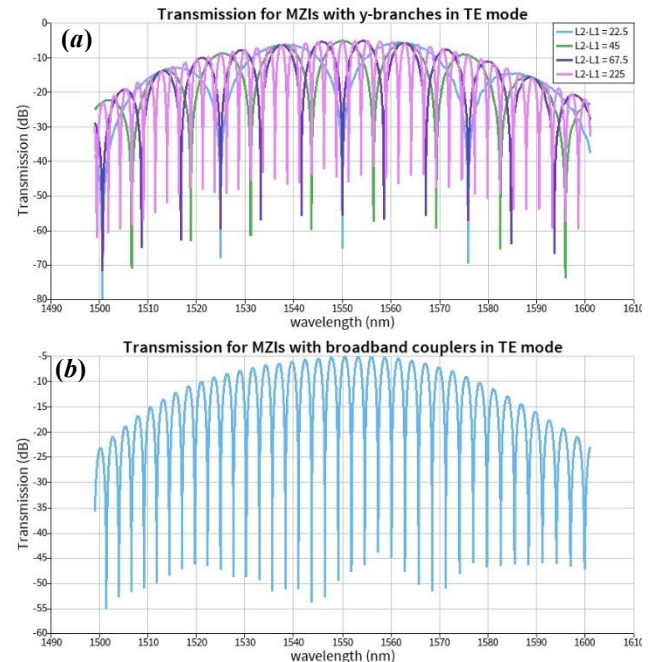


Fig. 3. Transmission of TE mode MZIs with a) y-branches and b) broadband couplers.

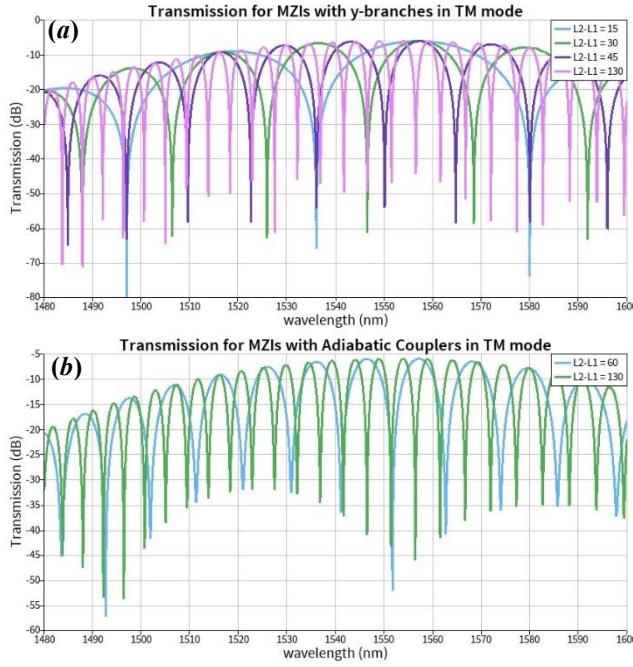


Fig. 4. Transmission of TM mode MZIs with a) y-branches and b) adiabatic couplers.

The loopback represents the losses accounted for the grating couplers and waveguides, which are added to the device performance. Fig. 5 shows the loopback effect applied by two TM grating couplers and a 156 μm waveguide, on the MZI with $\Delta L = 130$ nm. As shown in this figure, the grating couplers have applied a loss of around -20 dB (total loss of -21.133 dB). However, the device's behavior after loopback calibration is acceptable.

For a 500 nm TM waveguide, corner analysis shows that $3.5101 < n_g < 3.8679$ is acceptable. Evaluating the results for this TM MZI shows that n_g for the fabricated device is around 3.7627. This value, compared to the theoretical value of 3.72), lies between the corner analysis limits. Fig. 6 shows the data extracted from the experimental data by calculating the minimum peaks from the device's spectrum. The fitted value

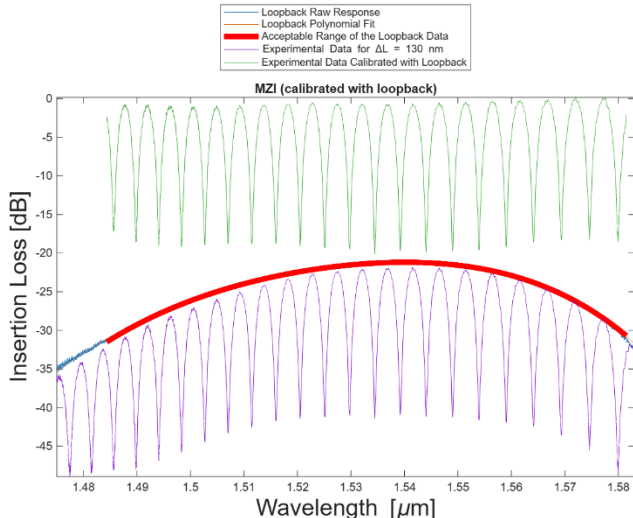


Fig. 5. Loopback calibration data, for a TM MZI. The correction is performed a 156 μm waveguide, for $\Delta L = 130$ nm.

(red line curve) is based on the FSR extracted from the minimums. Evaluated n_g is calculated by fitting the waveguide parameters; the TM waveguide response in the wavelength range of interest can be expressed as:

$$1.7757 - 1.2868 (\lambda - \lambda_0) + 1.8502 (\lambda - \lambda_0)^2 \quad (2)$$

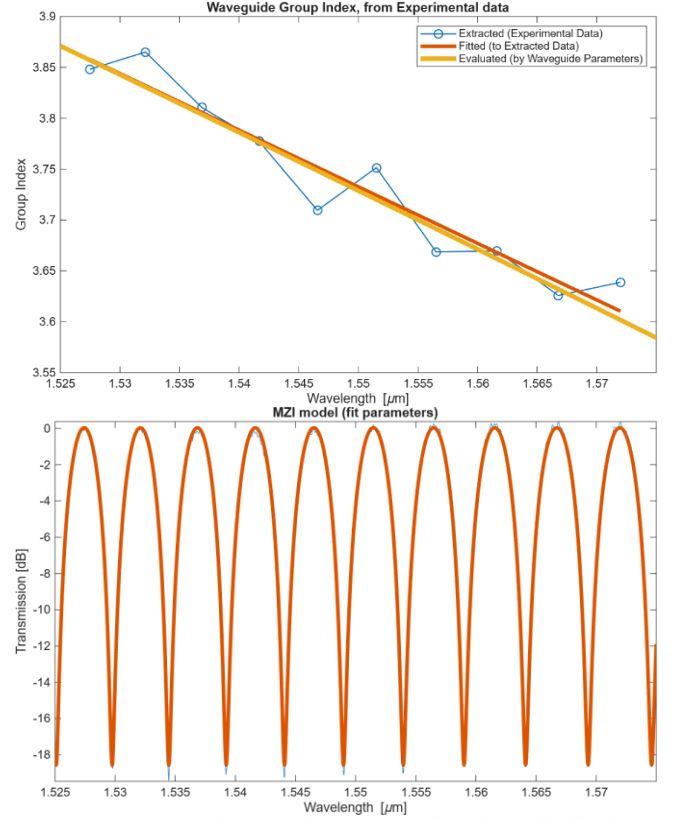


Fig. 6. a) Extracted n_g from the experimental results, and b) fitted MZI response after evaluating the waveguide parameters.

To summarize the results, we tabularized the waveguide parameters for TE- and TM-based MZIs, with y-branches. The parameters n_1 , n_2 , and n_3 , are the ones defined in Equation (1). We would like to emphasize that the values for the group index calculated here use the FSR values extracted from the dip points in the spectra.

Table 1. Waveguide parameters and at wavelength λ_{mean}

MZI	ΔL (μm)	n_1	n_2 (μm^{-1})	n_3 (μm^{-2})	λ_{mean} (μm)	n_g
TE	22.5*	FSR = 25.032 μm			1535	4.184
	45	2.3776	-1.2184	0.9825	1.518	4.227
	67.5	2.3948	-1.1669	-0.0755	1.532	4.1827
	225	2.3977	-1.1668	-0.0484	1.535	4.1884
TM	15*	FSR = 39.776			1.517	~3.857
	30	1.7565	-1.3642	2.0035	1.528	3.8406
	45	1.7615	-1.3139	1.8796	1.539	3.7839
	130	1.7638	-1.2945	1.8502	1.544	3.7627

* FSR is too big to achieve reasonable results

III. CONCLUSION

In this report, a 500 nm waveguide was simulated, and a compact model is presented based on Taylor's expansion for the TE model based on the simulations. The simulation results

are based on the compact models sourced from the Ebeam SiEPIC PDK package. The fabricated devices were used to evaluate the validity of the designs. The MZI was used to model waveguides and extract their parameters based on Taylor's expansion. The summarized results are acceptable, especially when they are based on the higher length difference, hence smaller FSRs.

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